Demonstrating Induction

Exam code: 0625 & 0972

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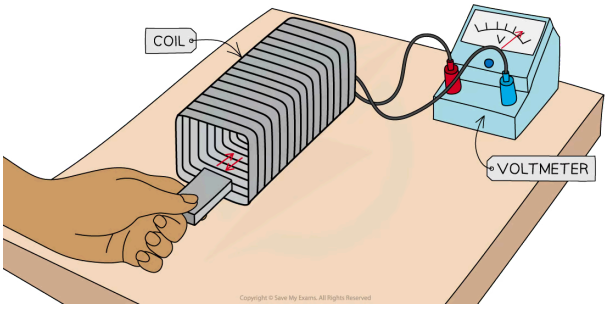
Demonstrating induction

- Electromagnetic induction is used in:
  - electrical generators which convert mechanical energy to electrical energy
  - transformers which are used in electrical power transmission
- The phenomenon of electromagnetic induction can be demonstrated using
  - a magnet and a coil
  - a wire and a U-shaped magnet

Experiment 1: moving a magnet through a coil

- When a coil is connected to a sensitive voltmeter, a bar magnet can be moved in and out of the coil to induce an e.m.f.



An e.m.f. is induced in a coil when a bar magnet is moved through it. This can be seen by connecting the coil to a voltmeter

- The expected results are...

1. When the bar magnet is stationary, the voltmeter shows a zero reading

- When the bar magnet is held still inside, or outside, the coil, there is no cutting of magnetic field lines
- As a result, **no e.m.f. is induced** in the coil

2. When the bar magnet is moved inside the coil, there is a reading on the voltmeter

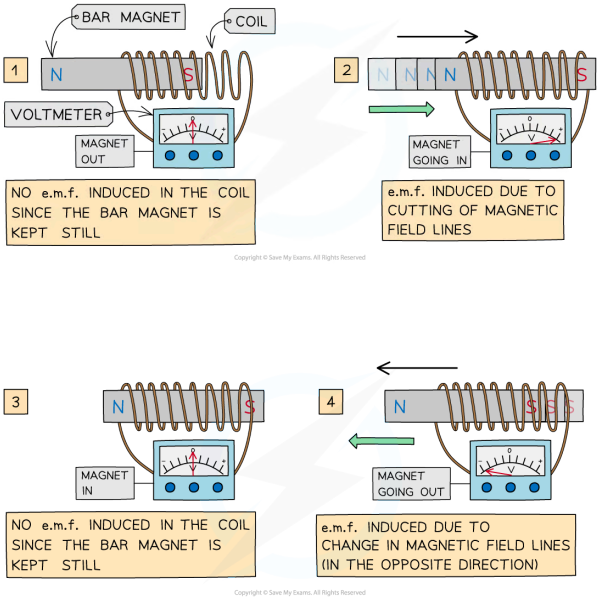
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

by the reading on the voltmeter

3. When the bar magnet is moved back out of the coil, there is a reading on the voltmeter with the opposite sign

- As the magnet changes direction, the direction of the current changes
- An e.m.f. is induced in the **opposite direction**, shown momentarily by the reading on the voltmeter with the opposite sign

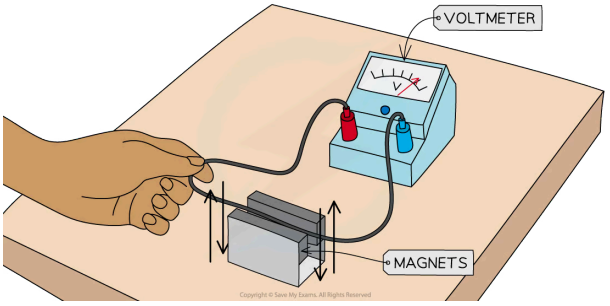


An e.m.f. is induced only when the bar magnet is moving through the coil

- Factors that will increase the induced e.m.f. are:
  - moving the magnet **faster** through the coil
  - adding **more turns** to the coil
  - increasing the **strength** of the bar magnet

Experiment 2: moving a wire through a magnet

- When a long wire is connected to a voltmeter and moved between two magnets, an e.m.f. is induced
- The pattern of a magnetic field in a wire can be investigated using this setup
  - **Note:** there is no current flowing through the wire to start with



- The expected results are...

1. When the wire is stationary, the voltmeter shows a zero reading

- When there is no relative motion between the wire and the magnetic field, no field lines are cut
- As a result, **no e.m.f. is induced** in the wire

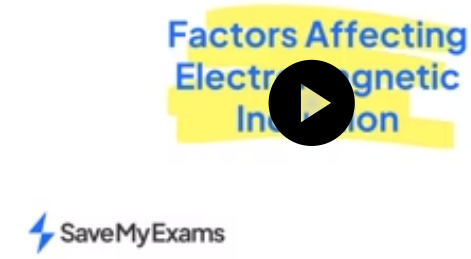
2. As the wire is moved between the magnetic poles, there is a reading on the voltmeter

- As the wire moves, it cuts the magnetic field lines of the magnet
- This induces an **e.m.f.** in the wire, shown momentarily by the reading on the voltmeter

3. When the wire is moved back out of the magnet, there is a reading on the voltmeter with the opposite sign

- As the wire changes direction, the direction of the current changes
- An e.m.f. is induced in the **opposite direction**, shown momentarily by the reading on the voltmeter with the opposite sign

- Factors that will increase the induced e.m.f. are:
  - increasing the **length** of the wire
  - moving the wire between the magnets **faster**
  - increasing the **strength** of the magnets



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Factors affecting electromagnetic induction

Factors affecting the magnitude of the induced e.m.f.

1. The speed at which the wire, coil or magnet is moved:

2. The number of turns on the coils in the wire:

- Increasing the number of turns on the coils in the same length of wire will increase the size of the induced emf
- Reducing the length of the wire but maintaining the same number of coils will also increase the size of the induced emf
- This is because each turn (loop) of wire in the coil cuts the magnetic field lines
- Therefore, the total induced e.m.f. increases with each additional turn (loop)

3. The size of the coils:

- Increasing the area of the coils will increase the size of the induced e.m.f.
- This is because there will be more wire to cut through the magnetic field lines

4. The strength of the magnetic field:

- Increasing the strength of the magnetic field will increase the size of the induced e.m.f.
- This is because there will be more magnetic field lines in a given area

Factors affecting the direction of the induced e.m.f.

1. The orientation of the poles of the magnet:

- Switching the poles of the magnet induces an e.m.f. in the opposite direction

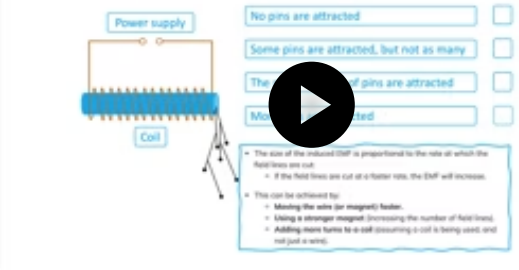
2. The direction in which the wire, coil or magnet is moved:

- Reversing the direction in which the wire, coil or magnet is moved induces an e.m.f. in the opposite direction

Examiner Tips and Tricks

When discussing factors affecting the size of an induced e.m.f., make sure to use the correct terminology:

- say "add more **turns** to the coil" instead of “add more coils”. This is because these statements do not mean the same thing
- say "a **stronger** magnet" instead of "a bigger magnet". This is because larger magnets are not necessarily stronger



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